

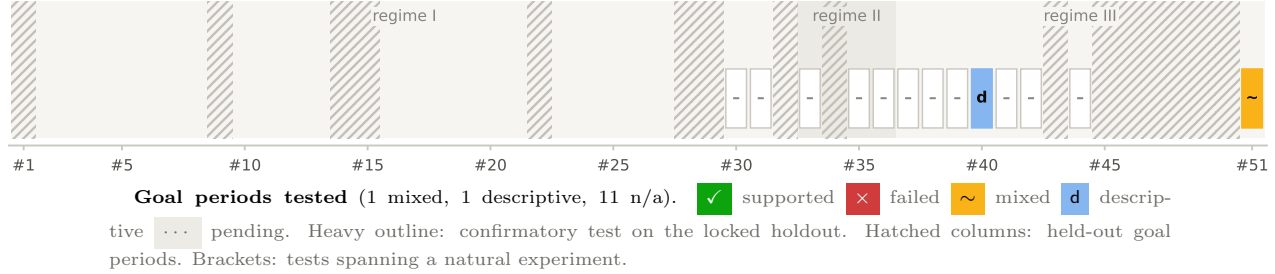
H135: Max-ent allocation as the steady state of a detailed-balance Potts walker: hop-rate ratios from max-ent occupancies

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 40% Faithfulness 0.5/5 Usefulness 1/5

Bottom line. In five own-role #51 attention units, hop destinations are not heat-bath on the max-ent occupancies ($\psi = 0.44$ [0.41, 0.48], habit $\rho = 1.9$ [1.4, 2.4], 5/5); zero net flux holds as written but is unpowered, so detailed balance is neither shown nor refuted.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



WHAT HAPPENED

HH378 says that H94's max-ent allocation is the steady state of a reversible kinetic Potts walker, so the hop-rate ratio between two projects equals their max-ent occupancy ratio. H135 rebuilds every agent's project hops (work host labels; attention labels) and refits H94's max-ent occupancies π per period unit on non-reserved data, then tests zero net flux between projects that live through the unit and the heat-bath destination law. **The round is inconclusive by the card's rule.** The strict co-alive rule leaves 0/62 unit-channels testable, and the zero-flux test has no power against the rival walkers. On the five #51 attention units testable under the card's 80% variant, **destinations are not heat-bath:** $\psi = 0.44$ [0.41, 0.48] instead of 1, with a habit bonus $\rho = 1.9$ nats.

PHYSICAL MODEL

Models 10 and 15. Agent i walks on the available projects with field $h_i(j) = \ln \pi_i(j)$, π_i the agent-conditional of H94's max-ent fit (margins, ownership, rooms). At each own call it updates with an aging hazard; the heat-bath update draws the new project from π_i . Detailed balance then gives zero net flux on every pair and destinations $\propto \pi_i$. Rivals: age drift, herding sink, cycle force, habit.

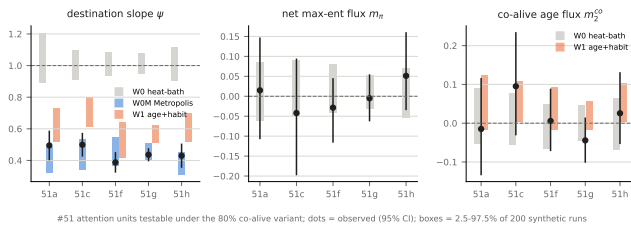
$$\frac{k_{ab}}{k_{ba}} = \frac{n_{ab}}{n_{ba}} \frac{T_b}{T_a} \stackrel{\text{DB}}{=} \frac{\pi_b}{\pi_a}, \quad m_\pi = \frac{\sum_{\text{pairs}} (n_{ab} - n_{ba}) \operatorname{sgn} \ln(\pi_b / \pi_a)}{\sum (n_{ab} + n_{ba})}$$

$$P(b \mid \text{hop from } a) \propto \exp[\psi \ln \pi_i(b) + \rho [b \text{ held}]], \quad \text{heat-bath: } \psi = 1, \rho = 0$$

Observables. Observed statistics per testable unit (#51 attention, 80% co-alive variant; dots with 95% CIs) against 200 synthetic runs on each unit's own skeleton (boxes, 2.5–97.5%). Left: the destination slope ψ sits far below the heat-bath band (gray) and inside the Metropolis band (blue) in 4/5 units. Middle and right: the net max-ent flux and the co-alive age flux sit inside the heat-bath band, but the synthetic shows this test cannot reject the rivals.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

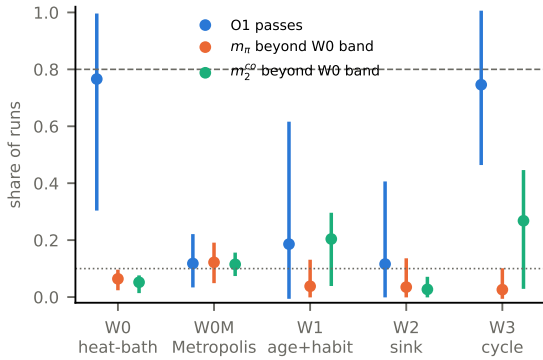
For an operator: the max-ent prices (ownership, rooms) set where agents spend their time, but they do not set where a moving agent goes next. A hopping agent responds to a project's max-ent weight with less than half the heat-bath exponent and prefers projects it has held before by about $e^{1.9} \approx 7\times$. To redirect agents, name the target explicitly; raising its general attractiveness moves hop destinations only weakly.



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RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
Precondition (strict co-alive)	0/62 unit-channels; 5 under the 80% variant	untestable
S1 O1 is an identity	passes in up to 1.00 of cycle-world runs	holds
P1 rate-ratio slope (HH)	1.11–1.60, $r > 0$ in 5/5	consistent only
P2 zero net flux m_π	5/5 inside band; RE -0.002 $[-0.04, 0.04]$	unpowered
P3 zero co-alive age flux	4/5 inside	unpowered
P4 heat-bath destinations	0/5 (ψ 0.39–0.50; ρ 1.2–2.6)	failed
N1 #38, N2 #44	too few co-alive hops	untestable
N3 #40 hub flux days 1–2	0.00 work, 0.04 attention	unpowered



Second figure. Synthetic validation on the five testable skeletons (200 runs per world; dots = mean, bars = range over skeletons). O1 passes as often in the cycle world as in the reversible one, so it is an occupancy identity. The flux statistics leave the heat-bath band in at most 0.29 of age-drift and sink runs: a fixed age preference and symmetric herding are themselves reversible on living projects. Only a cycle force produces flux.

GOAL PERIODS IN DETAIL

G40 d descriptive
G51 $\sim$ mixed
G30 – n/a
G31 – n/a
G33 – n/a
G35 – n/a
G36 – n/a
G37 – n/a
G38 – n/a
G39 – n/a
+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	1
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Only own-role #51 attention units are testable, and only under the 80% co-alive variant; attention labels are artifact mentions, not work. The origin $\times$ destination LR test was recalibrated after the synthetic (its χ^2 form rejects 0.76–1.00 of heat-bath runs). $\psi < 1$ may partly be attenuation from noise in π_i (cross-fit not run). The N3 hub counts were seen before the band. Owners come from non-reserved commits (37 repos differ from H94).

ROUND 2

Build a flux test with power: a time-varying or nonreciprocal field (flux after an owner arrives; H137's one-way pairs). Fit the destination law on all hops in every unit and compare ψ across own-role and shared weeks. Cross-fit π on odd and even days.

- **H135-R1** A flux test with power.
- **H135-R2** Destination law.
- **H135-R3** Cross-fit π (odd/even days) to check that $\psi \neq 1$ is not attenuation from noise in π_i .

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